Presenter script: three operator cases

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80 slides / 180 teaching minutes / one notebook / three exercises

Use Say for the spoken explanation and Do / discuss for demonstrations and audience prompts. Timings include code execution, pair work and discussion; add breaks outside the 180-minute teaching schedule.

Start with the twelve-minute Stanford concept block, then seven minutes of Colab setup. Follow one Classroom Labs notebook in order through L0-L12, including L10B after L10. Leave both live-tool switches False for the standard route.

Python calculates fresh numerical results. L10B replays saved Llama replies and their original tool results. The recorded studies are separate from learner edits. L12 is a simulated readback and equipment_dispatch remains False.

Session route

1. Stanford concepts and Colab setup: slides 1-15, 00:00:00-00:19:00.

2. Voltage support: slides 16-37, 00:19:00-01:08:00.

3. Service restoration: slides 38-53, 01:08:00-01:43:30.

4. Cyber response: slides 54-69, 01:43:30-02:18:30.

5. Llama replay and operator review: slides 70-80, 02:18:30-03:00:00.

Exercises: slides 25, 44 and 59. Give each four minutes; reset the input afterward.

Open Classroom Labs in Colab

Open presenter view

Open the searchable web script

1. Stanford concepts and Colab setup

Slide 1. Hands-on: Three Operator Cases

00:00:00-00:00:30 / 0.5 minutes

Say: Today we will use one notebook for three operator studies. We will connect Stanford ideas to the cases, run calculations, change inputs, inspect saved Llama choices and explain what a person must still review.

Do / discuss: Have slides, Classroom Labs and the Participant Lab Guide ready. Pair participants. Allow breaks outside the 180 teaching minutes.

Open slide 1

Slide 2. What is agentic AI?

00:00:30-00:01:30 / 1 minutes

Say: An LLM agent works toward a goal by choosing a permitted tool, reading its result and choosing the next step. The application keeps the tool rules fixed. The numerical tool supplies the physical evidence.

Do / discuss: Ask which part supplies a physical result. Expected: the numerical tool. Ordinary programs can also contain loops.

Open slide 2

Slide 3. Stanford: tool calls become a workflow

00:01:30-00:02:30 / 1 minutes

Say: These original Stanford panels distinguish a model-selected function call from a workflow connecting several steps. We will inspect the requests and returned results, so the coordinating role is visible.

Do / discuss: Point to the original function-calling and workflow panels. Follow the source link for lecture p.2.

Read the Stanford lecture, p.2

Open slide 3

Slide 4. Stanford example: check before you write

00:02:30-00:03:45 / 1.25 minutes

Say: WikiChat separates retrieval, claim checking and writing. We borrow the checking habit for grid studies: the summary should follow the evidence returned by a tool.

Do / discuss: Point to fact checking before Draft. Ask what numerical evidence a grid proposal would need.

Read the Stanford lecture, p.19

Open slide 4

Slide 5. Voltage: a proposal needs a numerical check

00:03:45-00:05:15 / 1.5 minutes

Say: In L2, Python calculates a proposal and checks it. This reference plan uses about 73.62 percent of allowed current. Also review voltage and model coverage. A separate saved Llama episode later shows tool selection.

Do / discuss: Preview L2 now; run it after setup, L0 and L1. The bar is a reference result, separate from the P3 replay.

Open slide 5

Slide 6. Stanford example: tools need clear rules

00:05:15-00:06:30 / 1.25 minutes

Say: Genie Worksheets emphasizes a complete task description, beyond listing functions. Our Python rules state the inputs, limits and stop conditions. A model explanation cannot silently change them.

Do / discuss: Ask which task rules must remain fixed when comparing restoration plans.

Read the Stanford lecture, p.40

Open slide 6

Slide 7. Restoration: the service goal shapes the plan

00:06:30-00:08:00 / 1.5 minutes

Say: The service goal changes the preferred restoration plan. Priority weighting gives 130 total kW with 90 critical kW. Maximizing total gives 140 total kW with only 30 critical kW. The network and capacities stay the same.

Do / discuss: Compare the cyan critical-service segments. The varied tie is 40 kW, another tie stays at 100 kW, and totals include upstream load.

Open slide 7

Slide 8. Stanford: language and decision checks

00:08:00-00:09:15 / 1.25 minutes

Say: Stanford separates language understanding from a solver that checks a formal decision model. Our grid adaptation separates Llama coordination, Python numerical checks and operator review. The clinical-source example supplies an analogy.

Do / discuss: Explain SMT briefly as logical constraints and MaxSAT as supporting decision-changing conditions. Our lab uses numerical checks and no SMT guarantee.

Read the Stanford lecture, p.53

Open slide 8

Slide 9. Cyber: a clear answer still needs evidence

00:09:15-00:10:45 / 1.5 minutes

Say: L8 asks whether readings fit the model. A small mismatch can hide a wrong estimate. L9 asks a separate recovery question that requires an explicit trust premise and passing numerical checks.

Do / discuss: Ask why a clear explanation cannot establish sensor trust.

Open slide 9

Slide 10. Ask four questions about a study result

00:10:45-00:12:00 / 1.25 minutes

Say: Use four review questions: which result supports the claim, which assumptions were used, were the rules kept, and what happens next? These questions apply to all three labs.

Do / discuss: Collect one example of a missing check, then start Colab setup.

Read the Stanford lecture, p.52

Open slide 10

Slide 11. One notebook covers all three cases

00:12:00-00:13:00 / 1 minutes

Say: Only Classroom Labs is required today. Keep that notebook beside the slides. It contains all three cases and the L10B replay; the graph and trace notebooks are companion resources.

Do / discuss: Open Classroom Labs and save a copy. Keep the same notebook open for all three cases.

Open slide 11

Slide 12. Get ready in four small steps

00:13:00-00:14:30 / 1.5 minutes

Say: Colab supplies a temporary computer. Save your copy, connect a CPU runtime and run setup until Ready. The default route needs no GPU, API key or model download.

Do / discuss: Connect a CPU runtime and run the first setup cell. Wait for Ready. After a reset, rerun setup and the required earlier cells.

Open slide 12

Slide 13. Run a cell, then read what it returns

00:14:30-00:16:00 / 1.5 minutes

Say: Find the input, play button and returned output. Editing a number does not change an old output until the whole cell runs again. Follow the named L lesson cells in order.

Do / discuss: Locate input, play button and output. Run the complete named L cell; slide code is often an excerpt.

Open slide 13

Slide 14. Run L0: choose the classroom route

00:16:00-00:17:00 / 1 minutes

Say: L0 selects the classroom route. Keep both switches False. Numerical calculations run now, the controller uses saved weights, and the model replies are recorded. The data-file assertion checks the working folder.

Do / discuss: Run L0 with RUN_OPENDSS=False and RUN_LOCAL_LLAMA=False. If the data check fails, rerun setup.

Notebook: run full L0; stable cell classroom-002, excerpt lines 11-14.

Open slide 14

Slide 15. Run L1: does the evidence still match?

00:17:00-00:19:00 / 2 minutes

Say: The unchanged request and command match the evidence. Changing either breaks that match. A fresh study is needed for the new request; stale evidence alone does not prove the changed case infeasible.

Do / discuss: Run L1. Read binding_pass as True, False, False. Ask what changed in each row.

Open slide 15

2. Voltage support

Slide 16. Voltage: open L2

00:19:00-00:21:00 / 2 minutes

Say: This synthetic feeder is our voltage teaching network. We will compare two control proposals, check modeled voltage and current limits, then compare their penalty scores. The default model has limited coverage.

Do / discuss: Locate L2 and identify this synthetic teaching feeder.

Open slide 16

Slide 17. MPC plans; SAC proposes

00:21:00-00:22:30 / 1.5 minutes

Say: MPC plans from a model and a short forecast. SAC proposes settings using saved learned weights. Both are tools inside the wider workflow, and the same numerical checks must test their proposals.

Do / discuss: Define MPC as model predictive control and SAC as Soft Actor-Critic. SAC uses saved weights without training again.

Open slide 17

Slide 18. A real agent changes its next step

00:22:30-00:24:00 / 1.5 minutes

Say: This separate saved P3 example makes the agentic sequence visible. A SAC check fails at about 111.97 percent of the allowance. Llama requests the MPC fallback, whose current check passes at about 99.60 percent. Human review remains pending.

Do / discuss: Point to the failed check between Llama choices. This is the saved P3 MV/LV study; the fixed comparator reaches the same numerical outcome.

Open slide 18

Slide 19. Control settings and limits

00:24:00-00:26:00 / 2 minutes

Say: Active and reactive settings share the inverter power limit. Network voltage and current must also pass. Here, 100 percent current means the configured allowance, which is a fraction of nominal rating.

Do / discuss: Explain per unit as a value divided by its base. At RATING_FRACTION=0.80, 100% current denotes the reduced allowance.

Open slide 19

Slide 20. A tool is a Python function in this lab

00:26:00-00:27:30 / 1.5 minutes

Say: A tool has defined inputs and returned evidence. Your L2 cell calls the voltage function directly. In the separate agent study, Python validates Llama's named choice before calling an allowed function.

Do / discuss: Locate lab.voltage_lab on the next slide. Distinguish a direct Python call from a model-selected request.

Open slide 20

Slide 21. Run L2 with the default inputs

00:27:30-00:29:30 / 2 minutes

Say: Run L2 at the defaults. It calculates an MPC plan and a proposal from the saved SAC policy, then checks both. Read feasibility and allowed-current use before comparing the current-step penalty.

Do / discuss: Set LOAD_SCALE=1.0 and RATING_FRACTION=0.80. Run all of L2; read feasible, max_allowed_current_pct and current_step_J.

Notebook: run full L2; stable cell classroom-007, excerpt lines 1-4.

Open slide 21

Slide 22. Read the voltage and current plots

00:29:30-00:32:30 / 3 minutes

Say: Read both plots together. The voltage curves are sorted separately, so equal ranks need not be the same location. Compare current use against the 100 percent line, using the output from the current run.

Do / discuss: Ask a pair to describe one voltage check and one current check. Use the current run's outputs.

Open slide 22

Slide 23. Check limits before comparing scores

00:32:30-00:34:30 / 2 minutes

Say: Both reference proposals pass. MPC uses about 73.62 percent of the current allowance; SAC uses about 51.68 percent. MPC has the lower penalty here. This single case does not establish a universal controller advantage.

Do / discuss: Read feasibility before penalty. Reference J: MPC about 0.03110, SAC about 0.43643.

Open slide 23

Slide 24. A rule and a tool result answer different questions

00:34:30-00:36:00 / 1.5 minutes

Say: A document can state a current limit, while a numerical tool calculates a proposal's current. Keep the rule, result and remaining checks together. This displayed P3 number is separate from your fresh L2 result.

Do / discuss: Keep the P3 value separate from the fresh L2 exercise. Ask which remaining checks the operator must review.

Open slide 24

Slide 25. Exercise 1: tighten the current allowance

00:36:00-00:40:00 / 4 minutes

Say: Predict the effect of tightening the current allowance. Change only that input, rerun L2 and report each method's feasibility, percentage and penalty. Explain what the default model covers.

Do / discuss: FOUR MINUTES: predict for 1 minute, edit/run for 2, report for 1. Change only RATING_FRACTION to 0.75; keep LOAD_SCALE=1.0.

Open slide 25

Slide 26. Change one number and explain the difference

00:40:00-00:44:30 / 4.5 minutes

Say: The same current becomes a larger share of a smaller allowance, so the percentage rises. Both proposals still pass this particular exercise. Explain the returned result and its scope rather than expecting a rejection.

Do / discuss: Debrief: reference current use becomes 78.527895% for MPC and 54.854158% for SAC. Both pass; default coverage is MV only.

Open slide 26

Slide 27. Reset L2 before reading old model choices

00:44:30-00:46:30 / 2 minutes

Say: Restore the defaults before the saved-response lessons. L3 through L5 use the original test_01 archive. Your edited L2 run does not rewrite that historical case.

Do / discuss: Restore LOAD_SCALE=1.0 and RATING_FRACTION=0.80, then rerun L2.

Notebook: run full L2; stable cell classroom-007, excerpt lines 1-2.

Open slide 27

Slide 28. What runs now and what was recorded

00:46:30-00:48:30 / 2 minutes

Say: A fresh calculation runs now. A replay displays an old reply with its original inputs. The closing control-room pilot is a teaching proposal. Keep those evidence labels visible.

Do / discuss: Ask which evidence mode applies to L2 and to the coming L3 replies.

Open slide 28

Slide 29. L3 opens a separate recorded case

00:48:30-00:50:30 / 2 minutes

Say: L3 opens saved responses for test_01. Read the requested operation, public reason and recorded time. Displaying these records makes no new language-model call.

Do / discuss: Run L3. Read the original test_01 identifier, requested operation and exact public reason.

Notebook: run full L3; stable cell classroom-010, excerpt lines 1-2.

Open slide 29

Slide 30. What did the models choose?

00:50:30-00:52:30 / 2 minutes

Say: Different wording can request the same numerical trial. Two recorded replies agree on reactive-power relaxation. Agreement selects a trial; a numerical check still has to evaluate it.

Do / discuss: Ask what evidence is required before accepting the selected trial for review.

Open slide 30

Slide 31. Run L4: check the old choice again

00:52:30-00:54:30 / 2 minutes

Say: L4 checks the selected archived trial again. It compares majority selection with a saved LinUCB selector. The engineering result comes from the new check, while the selection models remain fixed.

Do / discuss: Run L4 after L3. Read the table and separate the fixed selection methods from new numerical checks.

Notebook: run full L4; stable cell classroom-013, excerpt lines 5-9.

Open slide 31

Slide 32. Separate the old choice from the new check

00:54:30-00:56:30 / 2 minutes

Say: Both selectors choose the same relaxation trial in this reference case, and its fresh check passes. A learned selection score and physical feasibility answer different questions.

Do / discuss: Read operation, new_check_passes and current_step_J. Reference penalty is about 0.031082.

Open slide 32

Slide 33. Run L5: check what the model covers

00:56:30-00:58:30 / 2 minutes

Say: L5 asks what the original plan's model covers. It audits the archived test_01 MV-MPC plan, rather than L4's selected relaxation trial. Keep OpenDSS disabled on the default route.

Do / discuss: Keep RUN_OPENDSS=False. Run L5, read optional_mode and the checked model scope.

Notebook: run full L5; stable cell classroom-016, excerpt lines 1-4.

Open slide 33

Slide 34. A PASS covers only the model checked

00:58:30-01:00:30 / 2 minutes

Say: A passing result applies to the checked model and limits. An MV-only pass leaves missing evidence for a request requiring both MV and LV. Obtain the wider check or keep HOLD for that request.

Do / discuss: Ask which missing coverage prevents an MV-only pass from supporting a full MV/LV request.

Open slide 34

Slide 35. Voltage: Llama asks to check the saved controller

01:00:30-01:03:00 / 2.5 minutes

Say: Return to the separate saved P3 study. Llama requests the missing SAC check. OpenDSS reports two current violations and about 111.97 percent of allowed current. That tool result supports considering another permitted step.

Do / discuss: Identify VERIFY_SAC_FIRST and the recorded numerical failure. Replay the full source later in L10B.

Open slide 35

Slide 36. Voltage: Llama chooses the existing fallback plan

01:03:00-01:05:30 / 2.5 minutes

Say: An MPC fallback already exists in the saved case. Llama selects it for a check; Python and OpenDSS supply the passing result. The language model coordinates the request rather than solving the power flow.

Do / discuss: Identify USE_MPC_PLAN. Contrast Llama's requested step with the returned numerical evidence.

Open slide 36

Slide 37. Voltage: the checks pass and human review remains

01:05:30-01:08:00 / 2.5 minutes

Say: The numerical checks pass, while acknowledgment remains pending. The fixed workflow reached the same numerical result. Describe the coordination role and the missing human decision without claiming a numerical benefit from the model.

Do / discuss: Partner recap: one fresh calculation, one saved model choice and one unresolved operator review. Next, restoration.

Open slide 37

3. Service restoration

Slide 38. Restoration: open L6

01:08:00-01:09:00 / 1 minutes

Say: Restoration uses a separate six-node teaching graph. Keep the faulted branch unavailable. We will compare service goals, vary the tie capacity and inspect whether the evidence still matches its request.

Do / discuss: Locate L6 and keep the faulted branch unavailable.

Open slide 38

Slide 39. Choose the service goal before the plan

01:09:00-01:11:00 / 2 minutes

Say: The two plans follow different service goals under the same constraints. More total kW can come with less critical service. State the requested goal before deciding which plan is preferable.

Do / discuss: Ask pairs to state the service goal before choosing the preferred plan.

Open slide 39

Slide 40. Run L6: change the tie capacity

01:11:00-01:13:00 / 2 minutes

Say: L6 sweeps the tie capacity. The mixed-integer solver chooses whole loads under the graph rules. A separate exhaustive search checks the priority score on this small case.

Do / discuss: Run L6 with original capacities 40, 60, 80, 100 and 120 kW. Read served_kw, critical_kw and MILP_matches_exhaustive_check.

Notebook: run full L6; stable cell classroom-018, excerpt lines 1-5.

Open slide 40

Slide 41. Read both service plots

01:13:00-01:16:00 / 3 minutes

Say: Read total service on the left and critical service on the right. Coincident points mean the goals give the same result at that capacity. Compare both measures before explaining the tradeoff.

Do / discuss: Compare both plots at the same capacity. Ask where the service goals differ.

Open slide 41

Slide 42. The tradeoff at a 40 kW tie limit

01:16:00-01:18:00 / 2 minutes

Say: At the 40 kW varied-tie limit, priority weighting gives 130 total kW and 90 critical kW. The maximum-kW goal gives 140 total kW and 30 critical kW. Both use the same network and rules.

Do / discuss: Compare 130 total/90 critical with 140 total/30 critical kW. The goals share the same capacities and rules.

Open slide 42

Slide 43. How Python checks the plan

01:18:00-01:20:00 / 2 minutes

Say: Python checks source connection, capacities, power balance, the excluded fault and energized loops. Exhaustive enumeration checks the priority score. These graph checks leave AC physics and field switching order open.

Do / discuss: Ask which AC, protection and switching-order checks remain outside this formulation.

Open slide 43

Slide 44. Exercise 2: add more tie capacity

01:20:00-01:24:00 / 4 minutes

Say: Predict whether more tie capacity removes the tradeoff. Append 180 kW, keep 40 kW first and rerun L6. Compare total and critical service at those two capacities.

Do / discuss: FOUR MINUTES: predict for 1 minute, edit/run for 2, report for 1. Append 180. to CAPACITIES_KW and keep 40. first.

Open slide 44

Slide 45. Debrief: the tradeoff disappears here

01:24:00-01:27:00 / 3 minutes

Say: At 180 kW both goals give 250 total kW and 90 critical kW in this example. Priority weighting matters when constraints create a tradeoff; it does not improve every instance.

Do / discuss: Debrief: at 180 kW both methods serve 250 total and 90 critical kW. Explain why the constrained-case tradeoff disappears.

Open slide 45

Slide 46. Reset the capacity list, then run L7

01:27:00-01:29:00 / 2 minutes

Say: Restore the original capacity list before L7. The next cell binds the first 40 kW result to its request, then changes that request while keeping the old evidence.

Do / discuss: Restore the original capacity list and rerun L6. Run L7; it binds the first 40 kW record.

Notebook: run full L7; stable cell classroom-021, excerpt lines 1-3.

Open slide 46

Slide 47. Read the evidence-match test

01:29:00-01:31:00 / 2 minutes

Say: The original request matches. The changed request no longer matches the old evidence. The required next step is a fresh constrained study, rather than reusing an unsupported recommendation.

Do / discuss: Read original contract True and changed rating False. Request a fresh constrained solve.

Open slide 47

Slide 48. Changed inputs need a new check

01:31:00-01:33:00 / 2 minutes

Say: When inputs change, update the request, solve and check again, then bind the new evidence. An old correct calculation can still be irrelevant to a changed case.

Do / discuss: Ask which request fields and evidence must be refreshed after a change.

Open slide 48

Slide 49. Proposed agent role: keep the service goal

01:33:00-01:35:00 / 2 minutes

Say: The present cell runs Python: the operator supplies the service goal, and the optimizer chooses loads. A coordinating agent would request that study and explain the returned tradeoff while preserving the goal.

Do / discuss: Identify the current Python calculation and the proposed coordinating-agent responsibility.

Open slide 49

Slide 50. What this model does not check

01:35:00-01:36:00 / 1 minutes

Say: This transport model does not check AC voltage, reactive power or protection. An energized graph also does not supply a safe field switching sequence. Keep those missing checks explicit.

Do / discuss: Name a further study needed before a real operating decision.

Open slide 50

Slide 51. Restoration: Llama chooses a useful new trial

01:36:00-01:38:30 / 2.5 minutes

Say: In this separate recorded feeder study, Llama requests SHED_075. The tool returns about 1843.614 total kW and 474.040 critical kW, with a passing current check. The result establishes what that trial achieved.

Do / discuss: Read SHED_075, served/critical kW and the passing current check. This archive uses a separate feeder catalog; the fraction selects load objects, not percent kW.

Open slide 51

Slide 52. Restoration: the next trial crosses the current limit

01:38:30-01:41:00 / 2.5 minutes

Say: Llama requests SHED_065 next. The returned tie loading is about 111.009 percent, so the check rejects the trial. More served load cannot override a required current limit.

Do / discuss: Keep the previously passed candidate when SHED_065 fails its current check.

Open slide 52

Slide 53. Restoration: catch Llama's wrong ranking

01:41:00-01:43:30 / 2.5 minutes

Say: The final language ranking is wrong. Python compares the numerical costs, retains the lower-cost passed candidate E03 and rechecks it. A fluent explanation needs a guard grounded in the actual results.

Do / discuss: Find the wrong language order and numerical guard retaining E03. Contrast this saved feeder case with L6, then move to cyber.

Open slide 53

4. Cyber response

Slide 54. Cyber: open L8

01:43:30-01:44:30 / 1 minutes

Say: The cyber lab uses a separate four-bus DC fixture. We compare model fit with error against known simulation truth. That teaching truth is an offline comparison, not an extra authenticated operator measurement.

Do / discuss: Locate L8. Explain that the known truth is available for offline teaching comparison.

Open slide 54

Slide 55. Best fit and true state are different

01:44:30-01:46:30 / 2 minutes

Say: Weighted least squares finds the state that best fits the readings, weighting more precise measurements more strongly. A small residual measures consistency with the model; it does not establish authentic readings.

Do / discuss: Define WLS as weighted least squares. Ask what small residual establishes and what it leaves open.

Open slide 55

Slide 56. Run L8: compare two data attacks

01:46:30-01:48:30 / 2 minutes

Say: L8 compares an isolated biased reading with a coordinated shift fitting a different modeled state. This declared defensive fixture makes a detector limitation visible under stated model and access assumptions.

Do / discuss: Run the full default L8 cell. The defensive Hc example assumes knowledge of H and access to modeled unprotected channels.

Notebook: run full L8; stable cell classroom-023, excerpt lines 1-5.

Open slide 56

Slide 57. Read both cyber plots

01:48:30-01:51:30 / 3 minutes

Say: Follow each curve across both plots. Residual is on the left, while error against known teaching truth is on the right. The residual alarm can remain quiet while estimation error grows.

Do / discuss: Follow the same curve across residual and state-error plots.

Open slide 57

Slide 58. At scale 1, the alarm misses a state error

01:51:30-01:53:30 / 2 minutes

Say: The coordinated shift changes the estimated state while preserving the residual pattern of this linear fixture. At scale one the residual stays about 3.92 without an alarm, while state error is nonzero.

Do / discuss: Compare no residual alarm with nonzero state error at scale one.

Open slide 58

Slide 59. Exercise 3: add a small isolated bias

01:53:30-01:57:30 / 4 minutes

Say: Predict whether a small isolated bias triggers the alarm. Add scale 0.10, rerun L8, and read the alarm alongside the state error. Use both results to explain what no alarm means.

Do / discuss: FOUR MINUTES: predict for 1 minute, edit/run for 2, report for 1. Add .10 to ATTACK_SCALES and rerun L8.

Open slide 59

Slide 60. Debrief: no alarm does not prove truth

01:57:30-02:00:30 / 3 minutes

Say: The small isolated bias also stays below the chosen threshold. A quiet alarm alone cannot establish truth. Keep the claim tied to this fixture and its stated assumptions.

Do / discuss: Debrief: isolated-bias J about 10.143237 is below the 15.086272 threshold. Alarm=False despite nonzero error; coordinated J stays about 3.920222.

Open slide 60

Slide 61. Reset L8 before the recovery example

02:00:30-02:01:30 / 1 minutes

Say: Restore the default scale list and rerun L8. L9 uses a separate fixed recovery fixture; its correction does not follow the scale you just added.

Do / discuss: Restore the original ATTACK_SCALES tuple and rerun L8.

Notebook: run full L8; stable cell classroom-023, excerpt lines 1-3.

Open slide 61

Slide 62. L9 uses a fixed recovery case

02:01:30-02:03:30 / 2 minutes

Say: L9 compares a fixed recovery case with and without additional readings assumed trustworthy. We will separate that premise from the returned numerical result.

Do / discuss: Locate L9 and note that its fixed fixture does not follow the changed L8 sweep.

Open slide 62

Slide 63. Trusted readings are an assumption

02:03:30-02:05:30 / 2 minutes

Say: The three trusted angle readings are explicitly assumed. They supply three independent measurements for the three-angle model, with no residual redundancy left. This numerical calculation does not authenticate them.

Do / discuss: Ask who or what would establish the trusted-data premise outside this calculation.

Open slide 63

Slide 64. Run L9: compare HOLD and PROPOSE

02:05:30-02:07:30 / 2 minutes

Say: Run both L9 calls. Without trusted replacement data the result is HOLD. With the set assumed trusted and numerical checks passing, the result is PROPOSE. A proposal still needs review.

Do / discuss: Run all of L9, including its assertions. Read status, u_pu and offline_actual_violations.

Notebook: run full L9; stable cell classroom-026, excerpt lines 1-2.

Open slide 64

Slide 65. Read the two decision states

02:07:30-02:09:30 / 2 minutes

Say: The reference proposed injection is about 0.121694 per unit, with zero recorded offline model violations. This depends on the fixture and trust premise and does not authorize equipment dispatch.

Do / discuss: Read HOLD without trusted data and PROPOSE with the assumed trusted set. Reference u is about 0.121694 p.u.; offline toy violations are zero.

Open slide 65

Slide 66. Check the numerical correction

02:09:30-02:11:30 / 2 minutes

Say: The optimizer seeks the least-effort modeled correction under branch limits and a margin. Predicted flow12 is 0.17000 per unit; the offline truth check gives about 0.17046, below the 0.18000 rating.

Do / discuss: Compare predicted and offline flow checks. These use fixed DC sensitivities, rather than a full AC or dynamic model.

Open slide 66

Slide 67. What makes the recovery possible?

02:11:30-02:13:30 / 2 minutes

Say: The recovery depends on assumed trusted information and a constrained correction. Preserve both in the evidence packet so the operator can review the conditions supporting the proposal.

Do / discuss: Ask what must happen if the trust premise cannot be established. Retain HOLD.

Open slide 67

Slide 68. Cyber: Llama selects the tool already allowed

02:13:30-02:16:00 / 2.5 minutes

Say: In this separate saved cyber episode, the task already permits WLS_TRUSTED_REDISPATCH. Llama selects it and Python computes the correction. The model has not discovered or authenticated trusted sensors.

Do / discuss: Separate what the task already permits from what Llama chooses and Python computes.

Open slide 68

Slide 69. Cyber: a passing calculation still needs review

02:16:00-02:18:30 / 2.5 minutes

Say: The numerical calculation passes, while the overall result remains REVIEW and acknowledgment remains pending. Keep the trust premise and missing operating checks visible. The fixed comparison reaches the same numerical outcome.

Do / discuss: Partner recap: why are PASS and PROPOSE different from approved action? Then inspect the complete replays.

Open slide 69

5. Llama replay and operator review

Slide 70. Run L10 before the Llama walkthrough

02:18:30-02:21:30 / 3 minutes

Say: L10 opens a separate archived five-stage study. Read it for context, then continue to L10B for the readable saved Llama episodes. These records differ from your current classroom edits.

Do / discuss: Run L10 to inspect the separate archive. The later L11 switch stays False.

Open slide 70

Slide 71. Run L10B: watch Llama choose a tool

02:21:30-02:28:00 / 6.5 minutes

Say: Inspect the three saved episodes yourselves. Identify the supplied facts, allowed choice, exact public reason, returned tool result and next step. The replay makes no new model request or electrical solve.

Do / discuss: Run L10B for SCENARIO="voltage", then "restoration", then "cyber". Allow about 2 minutes per episode and swap runner/checker roles. Afterward run L11 with RUN_LOCAL_LLAMA=False.

Notebook: run full L10B; stable cell classroom-llama-walkthrough, excerpt lines 1-5.

Open slide 71

Slide 72. Finish L12: practice the readback

02:28:00-02:33:00 / 5 minutes

Say: L12 practices repeating the same recommendation ID and conditions. A wrong ID is rejected. The correct simulated acknowledgment still grants no equipment authority and does not resolve a scope gap.

Do / discuss: After L10B and L11, run all of L12. It returns to L4's selected test_01 voltage trial. Check wrong-ID rejection, simulated acknowledgment and equipment_dispatch=False.

Notebook: run full L12; stable cell classroom-033, excerpt lines 9-10.

Open slide 72

Slide 73. Read the two readback outcomes

02:33:00-02:36:00 / 3 minutes

Say: The full MV/LV request remains HOLD because its evidence lacks full coverage. An explicit MV teaching request can reach simulated readback. Each result stays tied to its actual request and recommendation ID.

Do / discuss: Explain the full-scope HOLD and the MV-only simulated readback. These do not complete the pending review of a separate old record.

Open slide 73

Slide 74. Llama chooses; tools calculate; people decide

02:36:00-02:37:00 / 1 minutes

Say: Llama chooses and explains permitted steps. Numerical software calculates and checks. The operator reviews the evidence and retains authority. These responsibilities shape the proposed advisory pilot.

Do / discuss: Assign one task to Llama, numerical software and the operator. Start the proposed advisory-pilot discussion.

Open slide 74

Slide 75. Start a control-room pilot with a clear study request

02:37:00-02:41:00 / 4 minutes

Say: Begin with a clear study request and an approved read-only snapshot containing time, topology version and limits. A stale topology copy needs refreshing before the study can support a current decision.

Do / discuss: Ask which snapshot fields must be refreshed after the illustrated topology mismatch. This is a teaching pilot, not a deployed integration.

Open slide 75

Slide 76. Give the assistant a small, controlled tool list

02:41:00-02:45:00 / 4 minutes

Say: Use a small approved tool list and a gateway that validates requests and logs results. Equipment commands remain in the utility's authorized operating process.

Do / discuss: Discuss one approved read-only function and one input-validation rule.

Open slide 76

Slide 77. Check three things before returning a proposal

02:45:00-02:49:00 / 4 minutes

Say: Before returning a proposal, check the evidence matches, all required limits pass and the requested scope is covered. A single PASS field cannot answer all three questions.

Do / discuss: Ask which of the three checks caught each workshop failure mode.

Open slide 77

Slide 78. Give the operator a short evidence packet

02:49:00-02:53:00 / 4 minutes

Say: Give the operator a concise evidence packet containing inputs, results, margins, rejected trials, scope and missing checks. The reviewer can request more study, reject the proposal or accept the study for the next review step.

Do / discuss: Ask pairs to draft a two-sentence handover for one case. Accepting a study packet does not operate equipment.

Open slide 78

Slide 79. Adopt it in stages and measure the benefit

02:53:00-02:57:00 / 4 minutes

Say: Adopt an assistant in stages, beginning with historical replay and read-only shadow studies. Measure correct work, missed checks, time and reviewer changes against a fixed workflow using the same tools.

Do / discuss: Choose one measurement that could demonstrate useful benefit against the fixed-workflow baseline.

Open slide 79

Slide 80. Save a workshop record others can repeat

02:57:00-03:00:00 / 3 minutes

Say: Save enough for another person to repeat the study. Explain the input, tool call, returned result, supporting checks and next operator decision. Mark each result as fresh, recorded or simulated.

Do / discuss: Collect an exit ticket: input, tool, result, checks and operator decision. Save the notebook and its changed inputs.

Open slide 80

Reading and source credits

The original Stanford figures and the grid teaching adaptations remain labeled. The classroom notebook uses custom Python, rather than WikiChat, Genie Worksheets or a theorem prover. Numerical model limits and trust premises remain explicit.

Complete teaching notes, numerical sources and figure credits

Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University. Fall 2026.

https://web.stanford.edu/class/cs224v/lectures_2026/l-introduction.pdf

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Harshit Joshi, Shicheng Liu, James Chen, Larsen Weigle, Monica Lam. Controllable and Reliable Knowledge-Intensive Task-Oriented Conversational Agents with Declarative Genie Worksheets. Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, Volume 1: Long Papers, pp.27264-27308. July 2025.

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Erik S., Barry Zhang. Building effective agents. Anthropic Engineering. 2024-12-19.

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